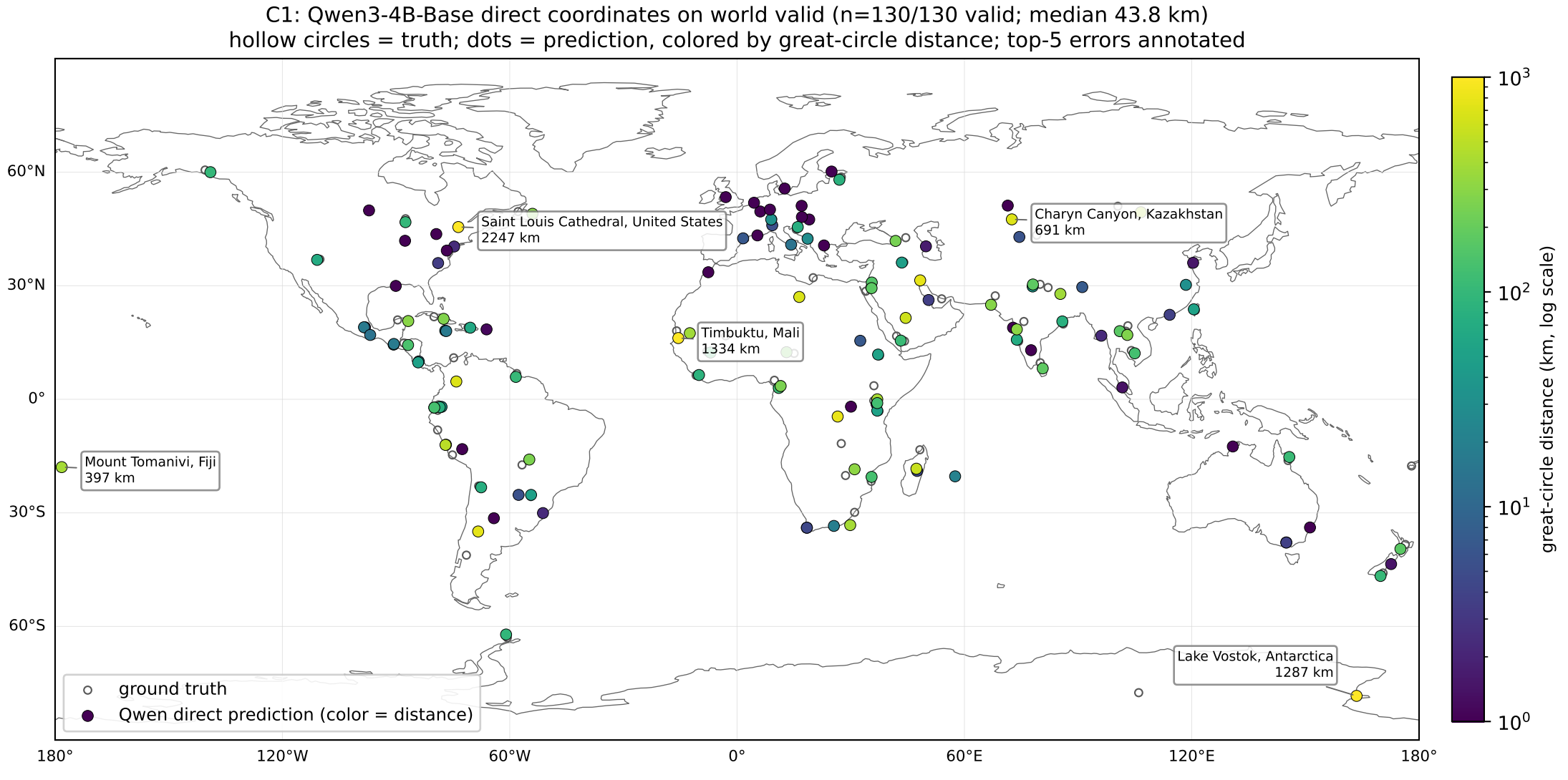

C1: Qwen3-4B-Base direct coordinates on world valid (n=130/130 valid; median 43.8 km)
hollow circles = truth; dots = prediction, colored by great-circle distance; top-5 errors annotated
Saint Louis Cathedral, United States
2247 km
Charyn Canyon, Kazakhstan
691 km
Timbuktu, Mali
1334 km
Mount Tomanivi, Fiji
397 km
Lake Vostok, Antarctica
1287 km
ground truth
Qwen direct prediction (color = distance)
great-circle distance (km, log scale)
10^3
10^2
10^1
10^0
60°N
30°N
0°
30°S
60°S
180°
120°W
60°W
0°
60°E
120°E
180°